PFSENSE SNORT

alert udp any 68 -> any 67 (msg:"kali linux detected";content:"kali";nocase;sid:1000001;)
alert tcp any any -> any any (msg:"Possible NMAP port scan";detection_filter:track by_src,count 20,seconds 60;sid:1000002;)
alert icmp any any -> any any (msg:"Large ping Packets detected";dsize:>128;sid:1000003;)
alert icmp any any -> any any (msg:"flooding detected";detection_filter: track by_src,count 30,seconds 60;sid:1000004)

sudo vim /etc/snort/rules/local.rules

sudo snort -A console -i ens37 -c /etc/snort/snort.conf -l /var/log/snort/

Install a web server on a Linux system. Install and configure Snort as IDS on this system. Add following rules to Snort to generate an alert for following threats.

1. If a ping packet with size greater than 128 bytes is received.
2. If from an outside IP address 30 ICMP packets are received within 1 min.
3. If from an outside single IP address more that 20 TCP SYN requests are received in 1 min to the web server
4. If a brute force attack is done on the SSH service.
5. If a kali Linux machine,bools in the network.
6. If a gmail website is accessed from the web server.
7. If the facebook website is accessed from the web server.
8. If the Instagram website is accessed from the web server.

alert icmp any any -> 192.168.74.201 any (msg:"larger message payload";dsize:>128;sid:1000001;)

alert icmp any any -> 192.168.74.201 any (msg:"Flood Detected ICMP";detection_filter:track by_src,count 30,seconds 60;sid:1000002;)

alert tcp any any -> 192.168.74.201 80 (msg:"Flood on WEB SERVER";flags:S;detection_filter:track by_src,count 20,seconds 60;sid:1000003;)

alert tcp any any -> 192.168.74.201 22 (msg:"ssh brute force attempt";flags:S;detection_filter:track by_src,count 5,seconds 60;sid:1000004;)

alert udp any 68 -> any 67 (msg:"kali linux detected";content:"kali"; nocase; sid:1000005;)

alert udp any any -> any any (msg:"Gmail login detected";content:"gmail"; nocase; sid:1000006;)

alert udp any any -> any any (msg:"Facebook Browsing detected";content:"facebook"; nocase; sid:1000007;)

alert udp any any -> any any (msg:"Insta accessed";content:"instagram"; nocase; sid:1000008;)

NGINX /etc/nginx/nginx.conf

```
# For more information on configuration, see:
#   * Official English Documentation: http://nginx.org/en/docs/
#   * Official Russian Documentation: http://nginx.org/ru/docs/

user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log;
pid /run/nginx.pid;

# Load dynamic modules. See /usr/share/doc/nginx/README.dynamic.
include /usr/share/nginx/modules/*.conf;

events {
    worker_connections 1024;
}

http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '
                      '$status $body_bytes_sent "$http_referer" '
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    tcp_nodelay         on;
    keepalive_timeout   65;
    types_hash_max_size 4096;
```

```
    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/ngx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

    upstream backend {
            server 192.168.25.25;
            server 192.168.25.35;
            #server 192.168.25.35 weight=5;
    }
    server {
        listen       80;
       # listen       [::]:80;
       # server_name  _;
       # root         /usr/share/nginx/html;

        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;

        error_page 404 /404.html;
        location = /404.html {
        }

        error_page 500 502 503 504 /50x.html;
        location = /50x.html {
        }
            #location = /exam {
            #         proxy_pass  http://192.168.25.25/;
            #}
            location = / {
                     proxy_pass  http://backend/;
            }
    }

# Settings for a TLS enabled server.
#
#    server {
#        listen       443 ssl http2;
#        listen       [::]:443 ssl http2;
#        server_name  _;
```

```
#        root         /usr/share/nginx/html;
#
#        ssl_certificate "/etc/pki/nginx/server.crt";
#        ssl_certificate_key "/etc/pki/nginx/private/server.key";
#        ssl_session_cache shared:SSL:1m;
#        ssl_session_timeout  10m;
#        ssl_ciphers PROFILE=SYSTEM;
#        ssl_prefer_server_ciphers on;
#
#        # Load configuration files for the default server block.
#        include /etc/nginx/default.d/*.conf;
#
#        error_page 404 /404.html;
#            location = /40x.html {
#        }
#
#        error_page 500 502 503 504 /50x.html;
#            location = /50x.html {
#        }
#    }

}
```

ipset icmp/ssh/http block :
iptables -I INPUT 1 -m set --match-set batman src -j DROP
iptables -I INPUT 2 -p icmp --icmp-type=echo-request -j SET --add-set batman src

----

Iptables :-

DMZ

sudo iptables -L

sudo iptables -t nat -L

sudo iptables -t nat -A POSTROUTING -s 192.168.25.0/24 -o ens160 -j MASQUERADE

sudo iptables -t nat -A POSTROUTING -s 192.168.35.0/24 -o ens160 -j MASQUERADE

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Enable Forwarding :-

cat /proc/sys/net/ipv4/ip_forward

sudo iptables -I FORWARD 1 -p udp -d 192.168.72.20 --dport=53 -j ACCEPT

sudo iptables -I FORWARD 1 -p udp -s 192.168.72.20 --sport=53 -j ACCEPT

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New chain for DMZ , Private :-

sudo iptables -N dmz_chain

sudo iptables -N lan_chain

sudo iptables -I FORWARD 3 -j dmz_chain

sudo iptables -I FORWARD 4 -j lan_chain

sudo iptables -A dmz_chain -p tcp -s 192.168.35.0/24 -m conntrack --ctstate NEW -j ACCEPT

sudo iptables -A dmz_chain -p tcp -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT sudo iptables -A dmz_chain -j RETURN

sudo iptables -A lan_chain -p tcp -s 192.168.25.0/24 -d www.google.com --dport=443 -m conntrack --ctstate NEW -j ACCEPT

sudo iptables -A lan_chain -p tcp -s 192.168.25.0/24 -d www.cdac.in --dport=443 -m conntrack --ctstate NEW -j ACCEPT

sudo iptables -A lan_chain -p tcp -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT

sudo iptables -A lan_chain -j RETURN

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Port Forwarding For Website Access :-

sudo iptables -t nat -A PREROUTING -p tcp -i ens160 --dport=80 -j DNAT --to-destination 192.168.35.10:80

sudo iptables -I dmz_chain 3 -p tcp -d 192.168.35.10 --dport=80 -m conntrack --ctstate NEW -j ACCEPT

Note: on machine 192.168.35.10, web server should be running
Sudo apt/dnf install httpd/apache2
Sudo vim /var/www/html/index.html

Option1: sudo ufw allow 80
Option2: sudo firewall-cmd --add-service=80 --permanent
Sudo firewall-cmd --reload

Browse 192.168.74.227- should show web page of server which in running on 192.168.35.10

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Ipset :-

Chai

ipset create chai hash:ip timeout 60
iptables -I INPUT 1 -m set --match-set chai src -j DROP
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iptables -I INPUT 2 -p icmp --icmp-type echo-request -j SET --add-set chai src
iptables -I INPUT 3 -p tcp --dport=22 -j SET --add-set chai src
iptables -I INPUT 4 -p tcp --dport=80 -j SET --add-set chai src

iptables -I INPUT 2 -p icmp --icmp-type echo-request -m length --length 156: -j SET --add-set chai src

iptables -R INPUT 2 -p icmp --icmp-type echo-request -m limit --limit 10/s -j SET --add-set chai src

ipset del chai 192.168.74.74

iptables -D INPUT 1
iptables destroy chai

Snort :-

Cd /etc/snort
Sudo vim snort.conf
--> ipvar HOME_NET 192.168.35.0/24
--> ipvar EXTERNAL_NET !$HOME_NET
--> var RULE_PATH /etc/snort/rules
--> var SO_RULE_PATH /etc/snort/so_rules
--> var PREPROC_RULE_PATH /etc/snort/preproc_rules
--> var WHITE_LIST_PATH /etc/snort/rules
--> var BLACK_LIST_PATH /etc/snort/rules

(make sure to enable this in file)

Cd /etc/snort/rules
Sudo vim local.rules

alert icmp any any -> any any (msg:"Test ping";sid:1000013;)
alert udp any any -> any 53 (msg:”LAN searching facebook”;content:”facebook”;nocase;sid:952145;)
alert tcp $EXTERNAL_NET any -> 192.168.74.227 22 (msg:"Test SSH";sid:1000003;)
alert icmp any any -> 192.168.74.227 any (msg:"Test big_data ping";dsize:>128;sid:1000004;)
alert icmp any any -> 192.168.74.227 any (msg:"Ping of Death";detection_filter:track by_src,count 20,seconds 60;sid:1000044;)
alert tcp any any -> 192.168.74.227 80 (msg:"flood web flow";flags:S;detection_filter:track by_src,count 20,seconds 60;sid:1000033;)

for logs :-
sudo snort -A console -i ens160 -c /etc/snort/snort.conf -l /var/log/snort/
(OR)
sudo cat /var/log/snort/alert.fast

NGINX

NOTE: set iptables before (remove REJECT rule from FORWARD and add MASQUERADE in NAT POSTROUTING)

Sudo setenforce 0

Sudo apt install nginx
Sudo systemctl status nginx

Cd /etc/nginx/
Sudo vim nginx.conf

```
 41    server {
 42        listen       80;
 43        #listen       [::]:80;
 44        #server_name  _;
 45        #root         /usr/share/nginx/html;
```

Load Balancer
Accord Line 37

```
 upstream backend {
        server 192.168.35.10;
        server 192.168.25.20 weight 5;
     }
         location / {
                          proxy_pass http://backend/;
         }
```

for different pages
Accord Line 58 for

```
          location / {
```

```
                    proxy_pass http://192.168.25.20/;
        }

        location /cdac{
                proxy_pass http://192.168.35.10/;
        }
```

Browser: 192.168.74.227/
Browser: 192.168.74.227/cdac

Note: ON client web server should be run

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nginx Loadbalancing + Reverse Proxy :-

Loadbalancing

```
include /etc/nginx/conf.d/*.conf;
upstream batman {
    server 192.168.25.10;
    server 192.168.25.20;
}
server {
    listen       80;
#     listen       [::]:80;
#     server_name  _;
#     root         /usr/share/nginx/html;

    # Load configuration files for the default server block.
    include /etc/nginx/default.d/*.conf;

    error_page 404 /404.html;
    location = /404.html {
    }
    location = / {
            proxy_pass http://batman/;
    }
    location = /cdac {
            proxy_pass http://192.168.25.10/;
```